

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Quantum Physics
4

L-T-P: 3-1-0

Credits:

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Need for Quantum Mechanics, Matter waves, Bohr-Sommerfeld quantization. Hilbert space, Observables and Hermitian Operators, Wave function and Schrodinger equation, probability interpretation. Eigen values and Expectation values, Uncertainty principle, Operator algebra and commutation relations. Ehrenfest theorem. (12)
One dimensional problems, Bound states and scattering states, Free particle, Particle in a box, Finite potential well, Potential barrier and Tunneling, Periodic potentials and bands, Harmonic oscillator. (14)
Three dimensional problems and angular momentum, Angular momentum algebra, Hydrogen atom. (16)

Reference Books:

1. Introduction to Quantum Mechanics: D. Griffiths
2. Principles of Quantum Mechanics : R. Shankar
3. Quantum Mechanics: P. M. Mathews & K. Venkatesan

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Need for Quantum Mechanics, Matter waves, Bohr-Sommerfeld quantization. Hilbert space, Observables and Hermitian Operators, Wave function and Schrodinger equation, probability interpretation. Eigen values and Expectation values, Uncertainty principle, Operator algebra and commutation relations. Ehrenfest theorem.	12
2.	One dimensional problems, Bound states and scattering states, Free particle, Particle in a box, Finite potential well, Potential barrier and Tunneling, Periodic potentials and bands, Harmonic oscillator.	14
3.	Three dimensional problems and angular momentum, Angular momentum algebra, Hydrogen atom.	16
Total number of hours		42